ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** energy	Main source.
2	**(b)** much sunlight	High albedo.
3	**(c)** carbon dioxide	CO2.
4	**(d)** atmosphere	Gases.
5	**(b)** absorb sunlight	Low albedo.
6	**(d)** carbon dioxide	CO2.
7	**(a)** geosphere	Solid.
8	**(b)** hotter than the poles	Direct rays.
9	**(d)** more CO2 in air	Fewer trees.
10	**(c)** hydrosphere	Water.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Geosphere and hydrosphere (1); Atmosphere and biosphere (1).	1+1
12	Change of water into vapour (2).	2
13	Fraction of sunlight reflected by a surface (2).	2
14	Trapping of heat by gases in the atmosphere (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fossil fuels (1); Deforestation (1); Agriculture and waste (1).	1+1+1
16	Save water (1); Save energy (1); Avoid plastics (1).	1+1+1
17	Absorb CO2 (1); Give oxygen (1); Rain (1).	1+1+1
	OR: Melting ice (1); Rising sea level (1); Extreme weather (1).	1+1+1
18	Solar cooker (1); Water heater (1); Solar cells (1).	1+1+1
	OR: Warming (1); Climate change (1); Sea-level rise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Geosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Interactions (1).	1+1+1+1+1
	OR: Source (1); Speed (1); Distribution (1); Reflection (1); Use (1).	1+1+1+1+1
20	Gases (1); Trap heat (1); Natural (1); Enhanced (1); Warming (1).	1+1+1+1+1
	OR: Cause (1); Gases (1); Effects (1); Control (1); Role of individuals (1).	1+1+1+1+1

Mistakes to avoid:

- Placing the ozone layer in the troposphere; it is in the stratosphere, and the weather happens in the troposphere.
- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** evaporation	Heat.
2	**(b)** sunlight	Solar energy.
3	**(d)** biosphere	Life.
4	**(b)** condensation	Cooling.
5	**(b)** renewable	Inexhaustible.
6	**(a)** greenhouse gas	Traps heat.
7	**(d)** precipitation	Falling water.
8	**(b)** Lifestyle for Environment	Eco-friendly.
9	**(b)** ultraviolet rays	UV.
10	**(b)** transpiration	Stomata.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Change of vapour into water droplets (2).	2
12	Carbon dioxide (1); Methane (1).	1+1
13	Rise in the Earth's average temperature (2).	2
14	Water falling as rain, snow or hail (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skin cancer (1); Cataract (1); Damage to crops (1).	1+1+1
16	Snow: high (1); Forest: low (1); Ocean: low (1).	1+1+1
17	t = d/v (1); 500 s (1); About 8 minutes (1).	1+1+1
	OR: Evaporation (1); Condensation (1); Precipitation (1).	1+1+1
18	Provides air (1); Protects from UV (1); Keeps Earth warm (1).	1+1+1
	OR: Plants (1); Animals (1); Microbes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Where (1); Function (1); CFCs (1); Effects (1); Protection (1).	1+1+1+1+1
	OR: Definition (1); Snow (1); Forest (1); Ocean (1); Effects (1).	1+1+1+1+1
20	Evaporation (1); Transpiration (1); Condensation (1); Precipitation (1); Runoff (1).	1+1+1+1+1
	OR: Meaning (1); Aim (1); Save energy (1); Save water (1); Reduce waste (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up the greenhouse effect with ozone depletion; greenhouse gases cause warming, CFCs cause ozone depletion.
- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
- Writing that plants release carbon dioxide in photosynthesis; photosynthesis absorbs CO_2 and releases O_2 (plants release CO_2 only in respiration).
- Not converting units in rainwater problems: $\text{area} \times \text{depth}$ gives m^3, which must be multiplied by 1000 to get litres.
- Thinking that the Earth's spheres work separately; they interact (for example evaporation links the hydrosphere to the atmosphere).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** energy	Electricity.
2	**(b)** stratosphere	Above.
3	**(a)** photosynthesis	Food making.
4	**(b)** water	Water.
5	**(d)** ozone layer	Chemicals.
6	**(c)** 78 per cent	Largest.
7	**(d)** avoided	Pollution.
8	**(c)** temperature	Average.
9	**(b)** pulses	Legumes.
10	**(b)** 3 x 10^8 m/s	c.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Conversion of atmospheric nitrogen into usable compounds (2).	2
12	Layer of ozone gas that absorbs UV rays (2).	2
13	Loss of water vapour by plants through stomata (2).	2
14	Lifestyle for Environment movement (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Respiration (1); Burning (1); Decay (1).	1+1+1
16	Equator: hot (1); Poles: cold (1); Sunlight angle (1).	1+1+1
17	Oceans (1); Rivers (1); Ice and groundwater (1).	1+1+1
	OR: Plant trees (1); Save energy (1); Use solar energy (1).	1+1+1
18	Fixation (1); Decay (1); Nitrification (1).	1+1+1
	OR: Rocks (1); Soil (1); Mountains (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Photosynthesis (1); Respiration (1); Decay (1); Burning (1); Ocean (1).	1+1+1+1+1
	OR: Renewable (1); Clean (1); Abundant in India (1); Uses (1); Limitations (1).	1+1+1+1+1
20	Fixation (1); Absorption (1); Food chain (1); Decay (1); Return (1).	1+1+1+1+1
	OR: Absorb CO2 (1); Give oxygen (1); Water cycle (1); Habitat (1); Soil (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that high albedo means a surface absorbs more heat; high albedo means more sunlight is reflected and less absorbed.
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